Deliverable 1
Structural Aliasing Analysis in Time Series Forecasting using Chronos

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This investigation formalizes the phenomenon of *structural aliasing* within patch-based time-series foundation models, focusing specifically on Amazon's Chronos-Bolt framework.

Intended Audience

This analysis is directly intended for researchers and practitioners utilizing large language models within the Chronos family for time-series analysis, representation learning, and forecasting, as well as individuals interested in characterizing the fundamental limitations, structural boundaries, and latent capabilities of these architectures.

This study is also explicitly directed toward researchers, engineers, and practitioners working in the field of energy management and domains directly related to this described infrastructure.

Purpose and Scope

This project formalises and empirically validates the phenomenon of *structural aliasing* in patch-based time-series foundation models, using Amazon Chronos-Bolt-Tiny as the target architecture.

Chronos-Bolt-Tiny operates with a fixed patch size of $P = 16$ and a default stride of $S = 16$, a context window of $T = 512$ tokens, and four encoder blocks followed by four decoder blocks terminating in a direct quantile projection head.

Ultimately, this study aims to provide a comprehensive understanding of the relationships between raw signal characteristics and the latent representations learned by the model, systematically identifying the critical links between sampling frequency (f_s), patch size (P), and stride (S) in order to propose actionable solutions to mitigate the phenomenon.

Layer-resolved probing is performed on the four encoder and decoder hidden states, plus at the utput probability distribution level. By mapping representations across these specific layers, the project isolates where task-relevant signal attributes are preserved or compromised under architectural synchronization constraints.

Stride is systematically varied within the range of $S \in [1, 16]$ to explore the explicit impact of patch geometry on aliasing effects.

The core objective is to challenge the baseline assumption that pre-trained temporal foundation models inherently preserve frequency attributes when processing Nyquist-compliant signals.

During the preparation of this work, the authors used AI-based tools, including GPT-5.5, GPT-5.4, Gemini 3.1 Pro, and Sonnet 4.6, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

Domain Description

To ground this abstract architectural evaluation, the study utilizes a specific physical domain as empirical reference layer. This reference setting models the solar energy produced by a photovoltaic array attached to an internal load, a local storage battery, and the national electricity grid. The explicit operational objective within this system is to evaluate production variations over short-term future horizons so that the system can dynamically modify the internal load, attach and detach the battery unit, with the aim of maintaining internal grid stability and enabling precise anomaly detection.

Within this infrastructure, a control AI agent is responsible for executing real-time tasks:

- **dynamic load-balancing**, which requires orchestrating the instantaneous routing of generated photovoltaic energy to meet shifting internal consumption demands;
- **battery dispatch**, which manages the continuous connection, isolation, or disconnection of the physical storage system to act as a buffer for intermittent generation;
- **state-monitoring and active anomaly detection**, parsing system-wide telemetry to differentiate between standard ambient fluctuations and hard sensor corruptions or hardware faults;
- **internal grid stability maintenance**, acting as a localized governor to minimize power distribution imbalances before they escape to the national grid interface.

Significance and Impact

This subject is of fundamental importance because it explore an unexamined systemic vulnerability native to patch-based temporal foundation architectures. By systematically characterizing the conditions under which structural aliasing emerges, this work will provide critical insights into the fundamental limitations of these models, directly informing the design of more robust architectures and training paradigms.

For professionals operating within this domain, the analysis could expose how algorithmic data-deletion anomalies can propagate through automated distribution pipelines. Evaluating how hidden token-space representation collapses can deceive a load-balancing AI agent into executing misaligned control choices, this work provides critical insights required to build high-reliability, fault-tolerant deployment models for smart-grid management systems.

The impact of this project could extend beyond theoretical understanding, as it has direct implications for the reliability and safety of real-world applications that depend on accurate time-series forecasting and representation learning, particularly in high-stakes domains such as energy management.

TODO

Add description of the point listed in the project outline including how we will achieve them, expected result and methodologies. Including a description about the method used to generate the signals.

Add references to the literature.

This document must address the following key questions:

- How will the project achieve its communication objectives?
- Descrizione dettagliata dei diversi punti da affrontare
- Quali sono i risultati attesi?
- Quali metodologie utilizzare per costruire dati sintetici?
- ?Analisi rappresentatività? (perdita di informazione e problematiche embedding (Probing MDL????))
- Analisi sui dati reali e risutlati attesi (Bayesian statistics)
- Implicazioni in termini di sicurezza sul dominio